JUNIOR LEVEL	PACKET TRACER	LAB 01

Lab 01

Device Configuration Fundamentals & IOS Navigation

Master Cisco IOS modes, basic device hardening, and essential show commands

IOS Navigation	Hostname / Banner	Password Security	SSH Config	Show Commands

CCNA Topic	1.1 – IOS Navigation, Device Hardening, Remote Management
Exam Objective	Configure device access security using local passwords
Difficulty	Junior
Simulator	Cisco Packet Tracer 8.x
Devices	Cisco 2911 Router + Cisco 2960 Switch + 2x PCs
Date Completed	2026-02-18

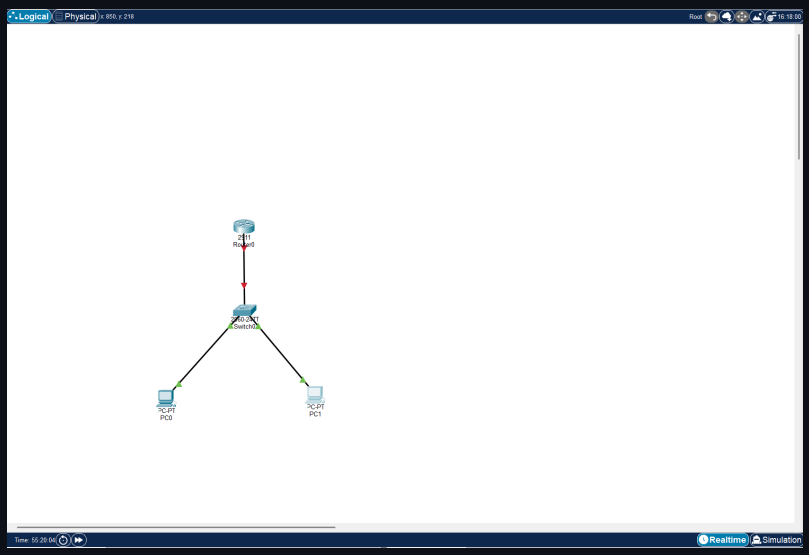

Figure 1 — Initial Packet Tracer topology (before configuration)

Lab Objective

Every CCNA exam and networking job starts here. In this lab we configured a Cisco 2911 router and 2960 switch from scratch in Cisco Packet Tracer: navigated all IOS modes, set hostnames and MOTD banners, secured console, VTY, and enable passwords, configured SSH for encrypted remote management, assigned IP addresses, and ran every essential show command. These skills form the foundation that every subsequent CCNA lab builds on.

Network Topology

Device	Model	Hostname	Role
Router	2911	R1	Gateway router — 192.168.1.1/24
Switch	2960	SW1	Access switch — VLAN1 192.168.1.2/24
PC	PC-PT	PC 1	Test workstation — 192.168.1.10/24
PC	PC-PT	PC 2	Test workstation — 192.168.1.11/24

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	N/A
SW1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PC 1	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC 2	Fa0	192.168.1.11	255.255.255.0	192.168.1.1

Cabling

`R1 GigabitEthernet0/0`	→	`SW1 GigabitEthernet0/1`	Copper Straight-Through
`PC 1 FastEthernet0`	→	`SW1 FastEthernet0/1`	Copper Straight-Through
`PC 2 FastEthernet0`	→	`SW1 FastEthernet0/2`	Copper Straight-Through

Step 1 — Router R1 Configuration

3.1 Hostname & MOTD Banner

The MOTD (Message of the Day) banner displays a legal warning to anyone connecting to the device. Never include the device name or model in the banner — attackers can use that information.

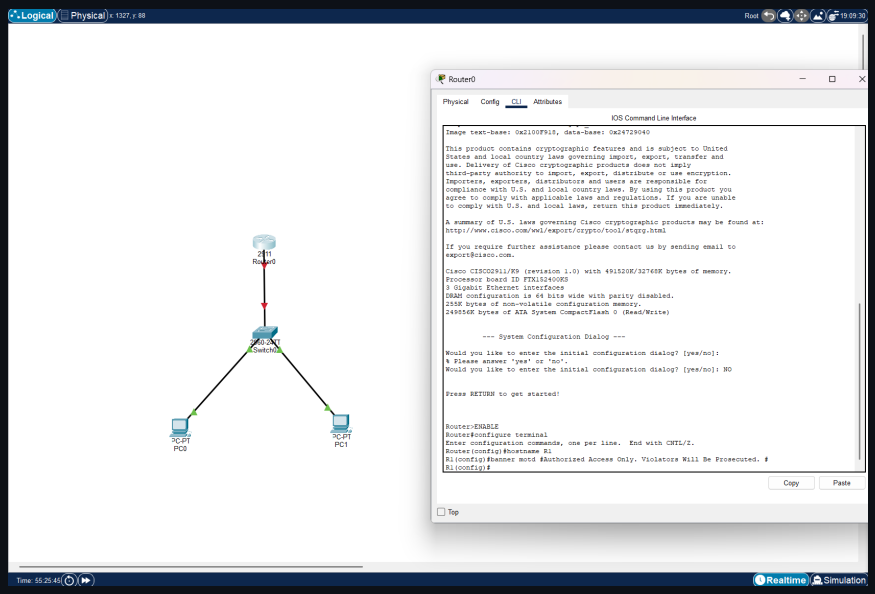

Figure 2 — Router CLI: hostname R1 and MOTD banner configured, entering global config mode

3.2 Password Security

Three layers of password protection are applied: the enable secret (MD5 encrypted), the console line password, and VTY passwords restricted to SSH only.

```
R1(config)#line console 0
R1(config-line)#password Console2020
R1(config-line)#login
R1(config-line)#logging synchronus
                                  ^
% Invalid input detected at '^' marker.

R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 5 0
R1(config-line)#
```

Figure 3 — Console line config: note the typo "synchronus" was caught and corrected immediately

Troubleshooting note: "logging synchronous" was initially mistyped as "logging synchronus". IOS returned *% Invalid input detected at '^'*. Retyped correctly and continued. Use Tab completion to avoid typos.

3.3 Router R1 — Full Configuration Commands

```
enable configure terminal hostname R1 banner motd # Authorized Access Only. Violators Will Be Prosecuted. #
enable secret Cisco2020 ! line console 0 password Console2020 login logging synchronous exec-timeout 5 0 exit
! line vty 0 4 login local transport input ssh exit ! service password-encryption username admin privilege 15
secret Admin2020 ! interface GigabitEthernet0/0 ip address 192.168.1.1 255.255.255.0 description ## Link to
SW1 ## no shutdown exit ! ip domain-name lab.local crypto key generate rsa general-keys modulus 2048 ip ssh
version 2 ip ssh time-out 60 ip ssh authentication-retries 3 ! end wr
```

Step 2 — Switch SW1 Configuration

The switch is configured with a management IP on VLAN 1, a default gateway pointing to R1, and SSH enabled identically to the router. Switches use VTY lines 0–15 (16 total) versus routers which use 0–4.

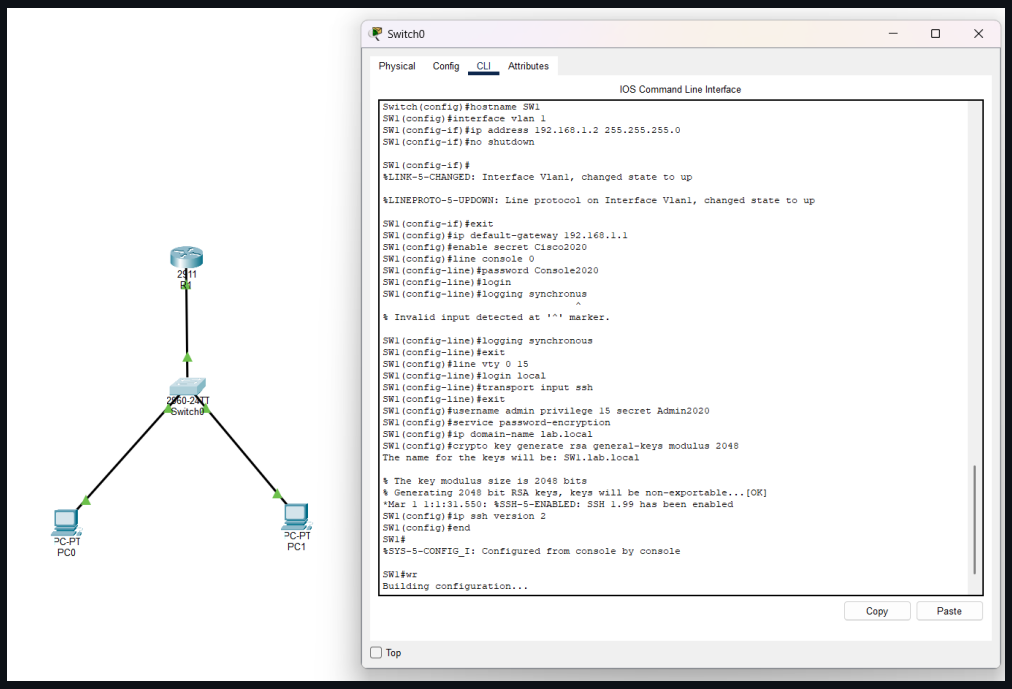

Figure 4 — SW1 CLI: full configuration including SSH key generation and wr (save)

SW1 — Key Configuration Highlights

Command	Purpose
interface vlan 1 / ip address 192.168.1.2 255.255.255.0	Assigns management IP to the switch
ip default-gateway 192.168.1.1	Allows switch to be reached from other subnets
line vty 0 15	Switches have 16 VTY lines vs 5 on routers
crypto key generate rsa general-keys modulus 2048	Generates RSA keys required for SSH
ip ssh version 2	Forces SSHv2 (more secure than v1)
wr	Saves running-config to startup-config (NVRAM)

Step 3 — PC IP Configuration

Both PCs are assigned static IP addresses via the Packet Tracer Desktop > IP Configuration GUI. No DHCP is used in this lab — all assignments are manual.

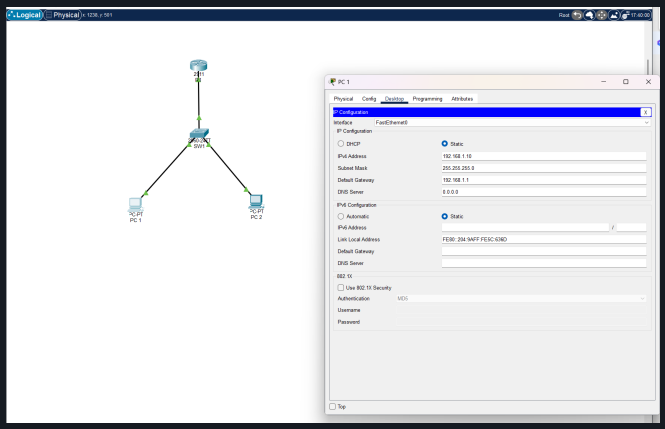

Figure 5 — PC 1: IP 192.168.1.10/24, GW 192.168.1.1

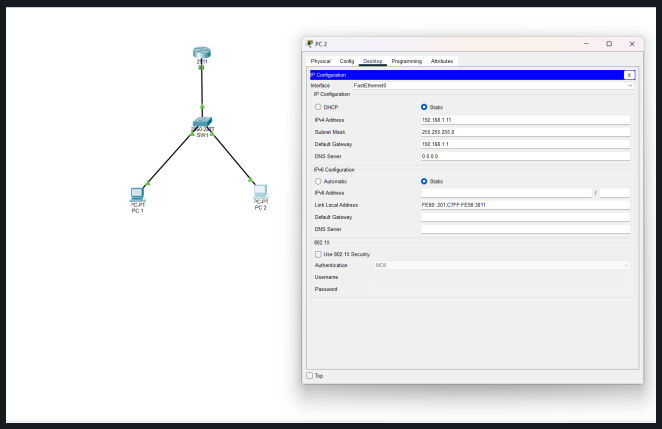

Figure 6 — PC 2: IP 192.168.1.11/24, GW 192.168.1.1

Setting	PC 1	PC 2
IP Address	192.168.1.10	192.168.1.11
Subnet Mask	255.255.255.0	255.255.255.0
Default Gateway	192.168.1.1	192.168.1.1

Step 4 — Connectivity Verification

All three ping tests were executed from PC 1. The first ping to SW1 (192.168.1.2) showed 25% packet loss — this is expected. The first ICMP packet triggers ARP resolution, which causes a timeout before the MAC address is cached. All subsequent pings succeed.

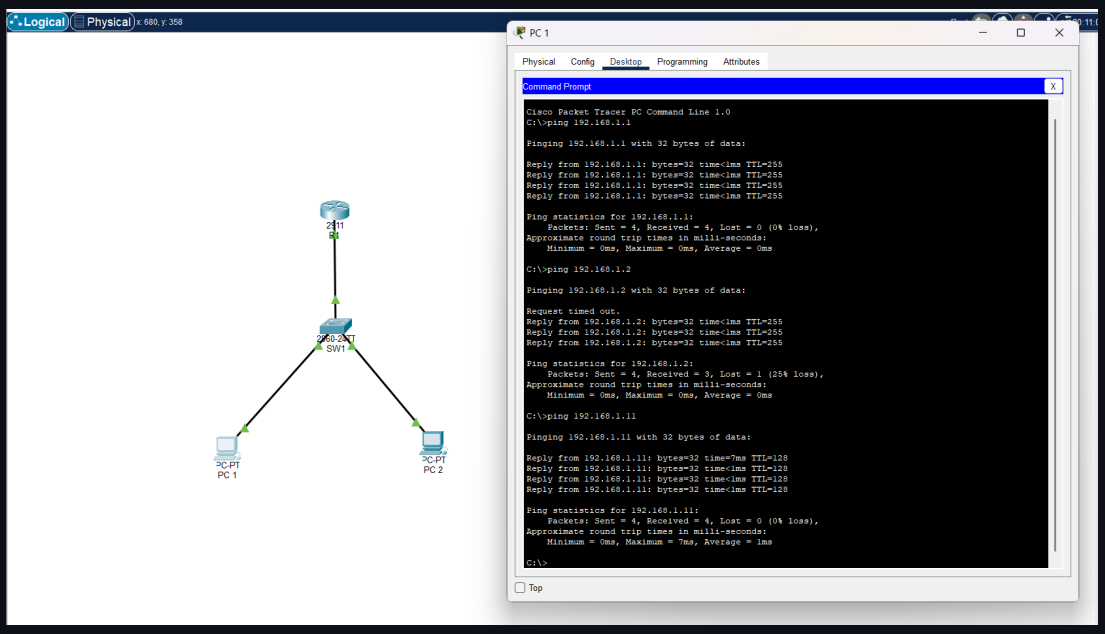

Figure 7 — PC 1 pinging R1 (192.168.1.1), SW1 (192.168.1.2), and PC 2 (192.168.1.11) — all successful

Ping Results Summary

Source	Destination	IP	Result	Notes
PC 1	R1	192.168.1.1	4/4 — 0% loss	Full success
PC 1	SW1	192.168.1.2	3/4 — 25% loss	ARP on first packet — normal
PC 1	PC 2	192.168.1.11	4/4 — 0% loss	Full success

Additional Verification — Ping Router Only

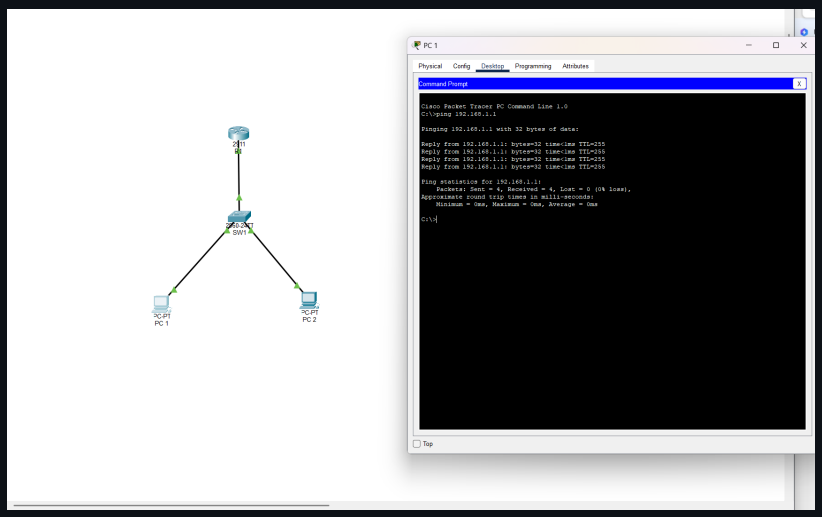

Figure 8 — Earlier ping test from PC 1 to R1 only, confirming router reachability

Troubleshooting Notes

One issue was encountered during the lab. It is documented here to demonstrate structured troubleshooting methodology.

Field	Details
Issue	logging synchronous typed as "logging synchronus"
Symptom	% Invalid input detected at '^' marker — IOS rejected the command
Root Cause	Typo — IOS does not auto-correct or accept partial misspellings
Fix	Retyped the correct command: logging synchronous
Prevention	Use Tab key to autocomplete commands. Use ? after partial input to see valid options

Pro Tip: The ? key is your best friend in IOS. Type part of a command then ? to see completions. Tab completes unambiguous commands. Ctrl+Z returns to Privileged EXEC from any config mode.

Essential Show Commands Reference

These commands were executed on R1 during the lab. Mastering them is critical for both the CCNA exam and real-world troubleshooting.

Command	When to Use	Exam Weight
`show ip interface brief`	FIRST command — quick status of all interfaces and IPs	HIGH
`show running-config`	Verify configuration changes took effect	HIGH
`show ip route`	Verify routing table entries and next hops	HIGH
`show version`	IOS version, uptime, hardware details	MEDIUM
`show interfaces G0/0`	Deep dive: errors, speed, duplex, counters	MEDIUM
`show arp`	IP-to-MAC mappings — verify reachability	MEDIUM
`show vlan brief`	VLAN assignments on switch ports (switches only)	HIGH
`show cdp neighbors`	Discover connected Cisco devices — topology mapping	MEDIUM
`show startup-config`	Verify saved configuration in NVRAM	MEDIUM
`show logging`	Review system log messages for errors	LOW

Key Takeaways

IOS Mode Navigation

IOS has five core modes: User EXEC → Privileged EXEC → Global Config → Interface/Line Config. Ctrl+Z always returns to Privileged EXEC from any config mode.

enable secret vs enable password

Always use enable secret — it uses MD5 encryption. enable password is plaintext unless service password-encryption is applied. If both are set, enable secret always wins.

service password-encryption

Encrypts all plaintext passwords in the config (Type 7 — reversible). Adds a layer of protection against shoulder-surfing but is NOT strong encryption.

SSH Requirements

SSH requires: (1) ip domain-name, (2) RSA key generation with minimum 2048-bit modulus, (3) ip ssh version 2, (4) local user account, (5) login local + transport input ssh on VTY.

Always Save Your Config

copy running-config startup-config (or wr) saves to NVRAM. Without this, all changes are lost on reload. This is one of the most common mistakes in real networking.

ARP Resolution Delay

The first ping to a new host often fails or times out — this is normal behaviour. The packet triggers ARP resolution which takes one round trip before the MAC is cached.

Switches vs Routers — VTY Lines

Routers have VTY lines 0–4 (5 total). Switches have VTY lines 0–15 (16 total). Configure all lines to ensure consistent behaviour.

Lab Validation Checklist

- ✓ R1 hostname set to R1
- ✓ MOTD banner with legal warning configured on R1
- ✓ enable secret encrypted password on R1 and SW1
- ✓ Console line password and logging synchronous on both devices
- ✓ VTY lines restricted to SSH only (transport input ssh)
- ✓ service password-encryption applied to both R1 and SW1
- ✓ R1 G0/0 IP address 192.168.1.1/24 — up/up
- ✓ SW1 VLAN 1 IP address 192.168.1.2/24 — up/up
- ✓ SW1 default gateway set to 192.168.1.1
- ✓ PC 1 static IP: 192.168.1.10/24, Gateway: 192.168.1.1
- ✓ PC 2 static IP: 192.168.1.11/24, Gateway: 192.168.1.1
- ✓ SSH with RSA 2048-bit keys and version 2 on R1 and SW1
- ✓ PC 1 ping R1 (192.168.1.1) — 0% loss (4/4)
- ✓ PC 1 ping SW1 (192.168.1.2) — ARP delay only, then success
- ✓ PC 1 ping PC 2 (192.168.1.11) — 0% loss (4/4)
- ✓ All IOS modes navigated and understood
- ✓ All essential show commands executed on R1

Lab 01 Complete — This is the foundation. Every CCNA lab builds on these skills.

CCNA Lab Series | Packet Tracer Edition | Junior Level